PU-I Biology | Chapter 2: Biological Classification

Worksheet: Lichens (Symbiotic Association)

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Name: ______________________________ Section: ___________ Date: _____________

SECTION A – Fill in the Blanks (1 mark each)

1. A lichen is a symbiotic association between a ______ and an alga.
2. The algal partner in a lichen is called the ______.
3. The fungal partner in a lichen is called the ______.
4. In a lichen, the alga prepares ______ for the fungus through photosynthesis.
5. In a lichen, the fungus absorbs water and ______ and provides shelter to the alga.
6. Lichens are considered excellent indicators of ______ pollution.
7. Lichens do not grow in ______ areas.
8. The fungal partners in lichens are typically drawn from Ascomycetes or ______.

SECTION B – One-Word / Very Short Answer (1 mark each)

1. What is the algal component of a lichen called? **Ans:** __________________________
2. What is the fungal component of a lichen called? **Ans:** __________________________
3. What type of relationship exists between the fungus and alga in a lichen? **Ans:** __________________________
4. Name one environmental factor that lichens are sensitive to. **Ans:** __________________________
5. Which partner in a lichen performs photosynthesis? **Ans:** __________________________

SECTION C – Match the Following (4 marks)

Column A	Column B
1. Phycobiont	a. Overall symbiotic structure
2. Mycobiont	b. Role played by lichens in ecology
3. Lichen	c. Algal partner
4. Pollution indicator	d. Fungal partner

Answer: 1-___, 2-___, 3-___, 4-___

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SECTION D – True or False (1 mark each)

1. Lichens are a symbiotic association between two different fungi. **[True / False]**
2. The mycobiont in a lichen is heterotrophic. **[True / False]**
3. Lichens thrive in heavily polluted industrial areas. **[True / False]**
4. The phycobiont in a lichen performs photosynthesis for the association. **[True / False]**

SECTION E – Short Answer (3 marks each)

1. Explain the roles of the phycobiont and mycobiont in a lichen.
2. Why are lichens considered good indicators of air pollution?
3. State the type of symbiotic relationship seen in lichens, with reasoning.

SECTION F – HOTS / Long Answer (5 marks each)

1. Discuss how the mutualistic relationship in lichens benefits both the fungal and algal partners.
2. Explain why the absence of lichens in an area might be ecologically significant to environmental scientists.

Teacher's Signature: _________________________________ **Date:** ______________

Teacher's Remarks: